

Answer ALL questions.

Write your answers in the spaces provided.

- 1 The photograph shows an Arencho plant (*Caylusea abyssinica*) that grows in East Africa.
Boiled leaves of Arencho are used as a traditional medicine. Extracts of this plant show antimicrobial properties.



(Source: © ChWeiss/Shutterstock)

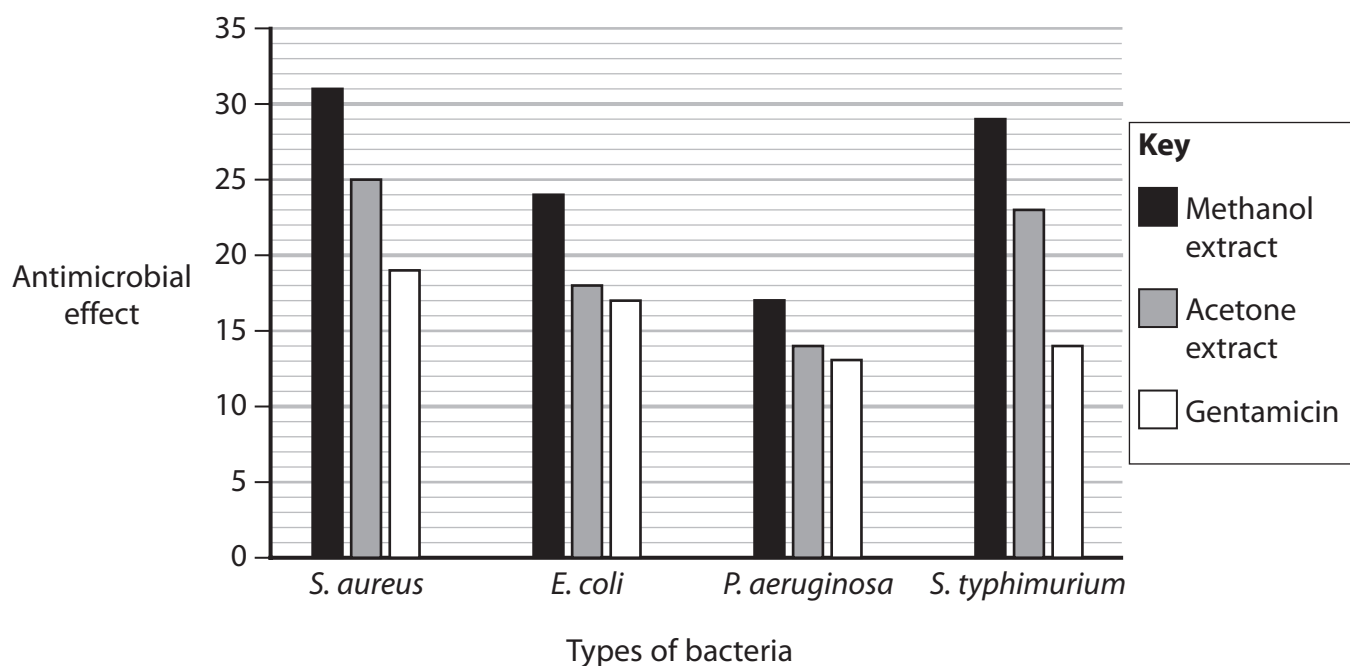
- (a) In an investigation, roots of *Caylusea abyssinica* were dried and powdered.

Extracts were made using two solvents: methanol and acetone.

The powdered root was shaken with each solvent and then filtered to produce the extracts.

These extracts were used to measure the antimicrobial properties on four types of bacteria. The antimicrobial properties of the antibiotic gentamicin were also measured.

The graph shows data from this investigation.



- (i) State a suitable dependent variable with units for this investigation.

(2)

Dependent variable

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Units

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- (ii) Describe a safe method, that uses your dependent variable, to obtain the results shown in the graph.

(5)

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- (iii) Calculate the percentage difference in the effects of the methanol extract and gentamicin on *S. typhimurium*.

Give your answer to **two** significant figures.

(2)

Answer %

- (iv) Comment on the antimicrobial effect of extracts of the Arencho plant.

(4)

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- (c) Plant extracts can be used to make new drugs for the treatment of infections. In trials to test these drugs, placebos and double-blind trials are used.

Explain the purpose of each of these.

(3)

placebos

double-blind trials

(Total for Question 1 = 20 marks)

